

SOLUTIONS

(1) (a) We have to compute $\iint_{\Omega} \lambda(x,y) dx dy = \iint_{\Omega} [(x-1/2)^2 + y^2] dx dy$.

Using polar coordinates $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} ; \begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \arctan(y/x) \end{cases}$

we have $(x-1/2)^2 + y^2 = (x^2 + y^2) - x + 1/4 = r^2 - r \cos \theta + 1/4$
and Ω is described by $0 \leq r \leq 1, 0 \leq \theta \leq 2\pi$. Hence using
the Jacobian of the change to polar coordinates: $dx dy = r dr d\theta$, we get:

$$\begin{aligned} M &= \iint_{\Omega} [(x-1/2)^2 + y^2] dx dy = \int_0^{2\pi} \int_0^1 [r^2 - r \cos \theta + 1/4] r dr d\theta \\ &= \int_0^{2\pi} \int_0^1 r^3 dr d\theta - \int_0^{2\pi} \int_0^1 r^2 \cos \theta dr d\theta + \int_0^{2\pi} \int_0^1 \frac{r}{4} dr d\theta \\ &= 2\pi \left[\frac{r^4}{4} \right]_0^1 - \underbrace{\left(\int_0^{2\pi} \cos \theta d\theta \right)}_0 \left(\int_0^1 r^2 dr \right) + 2\pi \left[\frac{r^2}{8} \right]_0^1 \\ &= \frac{2\pi}{4} + \frac{2\pi}{8} = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4} \end{aligned}$$

Hence $M = 3\pi/4$ is the mass of the disk.

(b) We have $x_M \cdot M = \iint_{\Omega} x \lambda(x,y) dx dy$
 $y_M \cdot M = \iint_{\Omega} y \lambda(x,y) dx dy$

$$\begin{aligned} \text{Hence } x_M M &= \int_0^{2\pi} \int_0^1 (r \cos \theta) [r^2 - r \cos \theta + 1/4] r dr d\theta \\ &= \int_0^{2\pi} \int_0^1 r^4 \cos \theta dr d\theta - \int_0^{2\pi} \int_0^1 r^3 \cos^2 \theta dr d\theta + \frac{1}{4} \int_0^{2\pi} \int_0^1 r^2 \cos \theta dr d\theta \end{aligned}$$

$\Rightarrow$

$$= \underbrace{\left(\int_0^{2\pi} \cos \theta d\theta \right)}_0 \left(\int_0^1 r^4 dr \right) - \left(\int_0^{2\pi} \cos^2 \theta d\theta \right) \left(\int_0^1 r^3 dr \right) + \frac{1}{4} \underbrace{\left(\int_0^{2\pi} \cos \theta d\theta \right)}_0 \left(\int_0^1 r^2 dr \right)$$

$$= - \left(\int_0^{2\pi} \cos^2 \theta d\theta \right) \cdot \left(\frac{1}{4} \right).$$

We need to compute $\int_0^{2\pi} \cos^2 \theta d\theta$. Recall the formula

$$\cos^2 \theta = \frac{1 + \cos(2\theta)}{2}. \text{ Hence } \int_0^{2\pi} \cos^2 \theta d\theta = \left[\frac{\theta}{2} \right]_0^{2\pi} + \left[\frac{\sin(2\theta)}{4} \right]_0^{2\pi}$$

$$= \pi + 0 = \pi.$$

We obtain $x_M \cdot M = -\frac{\pi}{4}$. Since $M = \frac{3\pi}{4}$, $\boxed{x_M = -\frac{1}{3}}$

Now we compute:

$$y_M M = \int_0^{2\pi} \int_0^1 y \, d(x,y) \, dx \, dy = \int_0^{2\pi} \int_0^1 (r \sin \theta) \left[r^2 - r \cos \theta + \frac{1}{4} \right] r \, dr \, d\theta$$

$$= \underbrace{\left(\int_0^{2\pi} \sin \theta d\theta \right)}_0 \left(\int_0^1 r^4 dr \right) - \left(\int_0^{2\pi} \sin \theta \cdot \cos \theta d\theta \right) \left(\int_0^1 r^3 dr \right)$$

$$+ \frac{1}{4} \underbrace{\left(\int_0^{2\pi} \sin \theta d\theta \right)}_0 \left(\int_0^1 r^2 dr \right) = - \left(\int_0^{2\pi} \sin \theta \cos \theta d\theta \right) \cdot \left(\frac{1}{4} \right)$$

$$\text{Now } \int_0^{2\pi} \sin \theta \cdot \cos \theta d\theta = \left(\int t \, dt \right) \Big|_{\substack{t = \sin \theta \\ dt = \cos \theta d\theta}} = \left[\frac{\sin^2 \theta}{2} \right]_0^{2\pi} = 0.$$

Hence $y_M \cdot M = 0$ and thus $y_M = \frac{0}{M} = 0$, $\boxed{y_M = 0}$

It follows that the center of mass of the plate is

$$\boxed{(x_M, y_M) = \left(-\frac{1}{3}, 0 \right)}$$

② (a) Standard form:

$$\min_{(x_1, x_2) \in \mathbb{R}^2} \begin{bmatrix} -8 \\ -4 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

subject to

$$\begin{bmatrix} 2 & 4 \\ -1 & 4 \\ 1 & -1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 8 \\ 4 \\ 2 \\ 0 \\ 0 \end{bmatrix}$$

that is, $c = \begin{bmatrix} -8 \\ -4 \end{bmatrix}$, $A = \begin{bmatrix} 2 & 4 \\ -1 & 4 \\ 1 & -1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}$, $b = \begin{bmatrix} 8 \\ 4 \\ 2 \\ 0 \\ 0 \end{bmatrix}$.

This is a convex optimization problem, because it is of the form

$\min f(x)$ subject to $g_i(x) \leq 0$ ($i=1 \rightarrow m$),
where $f(x)$ and $g_i(x) \rightarrow g_m(x)$ are convex functions, because
affine functions are convex (and concave).

(b) The Lagrangian is:

$$\mathcal{L}(x, \lambda) = c^T x + \lambda^T (Ax - b), \text{ where } \lambda = \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \\ \lambda_5 \end{bmatrix} \in \mathbb{R}^5$$

with $\lambda_1 \geq 0, \lambda_2 \geq 0, \lambda_3 \geq 0, \lambda_4 \geq 0, \lambda_5 \geq 0$.

that is, $\lambda \geq 0$, or $\lambda \in \mathbb{R}_+^5$.

We can write $\mathcal{L}(x, \lambda) = (c + A^T \lambda)^T x - \lambda^T b$.

Taking $\mathcal{L}_p(x) = \sup_{\lambda \in \mathbb{R}_+^5} \mathcal{L}(x, \lambda)$, we obtain the

Primal Problem

$$\min_{x \in \mathbb{R}^2} \mathcal{L}_p(x) = \min_{x \in S} f(x)$$

where $S = \{x \in \mathbb{R}^2 \mid Ax \leq b\}$.

Taking the dual :

$$\mathcal{L}_D(\lambda) = \min_{x \in \mathbb{R}^2} \mathcal{L}(x, \lambda)$$

we obtain the

Dual Problem

$$\max_{\lambda \in \mathbb{R}_+^5} \mathcal{L}_D(\lambda).$$

In our case, setting $\nabla_x \mathcal{L}(x, \lambda) = 0$, we get

$$\boxed{C + A^T \lambda = 0},$$

$$\max_{\lambda \in \mathbb{R}^5} -b^T \lambda$$

$$\text{subject to } \begin{cases} C + A^T \lambda = 0 \\ \lambda \geq 0 \end{cases}$$

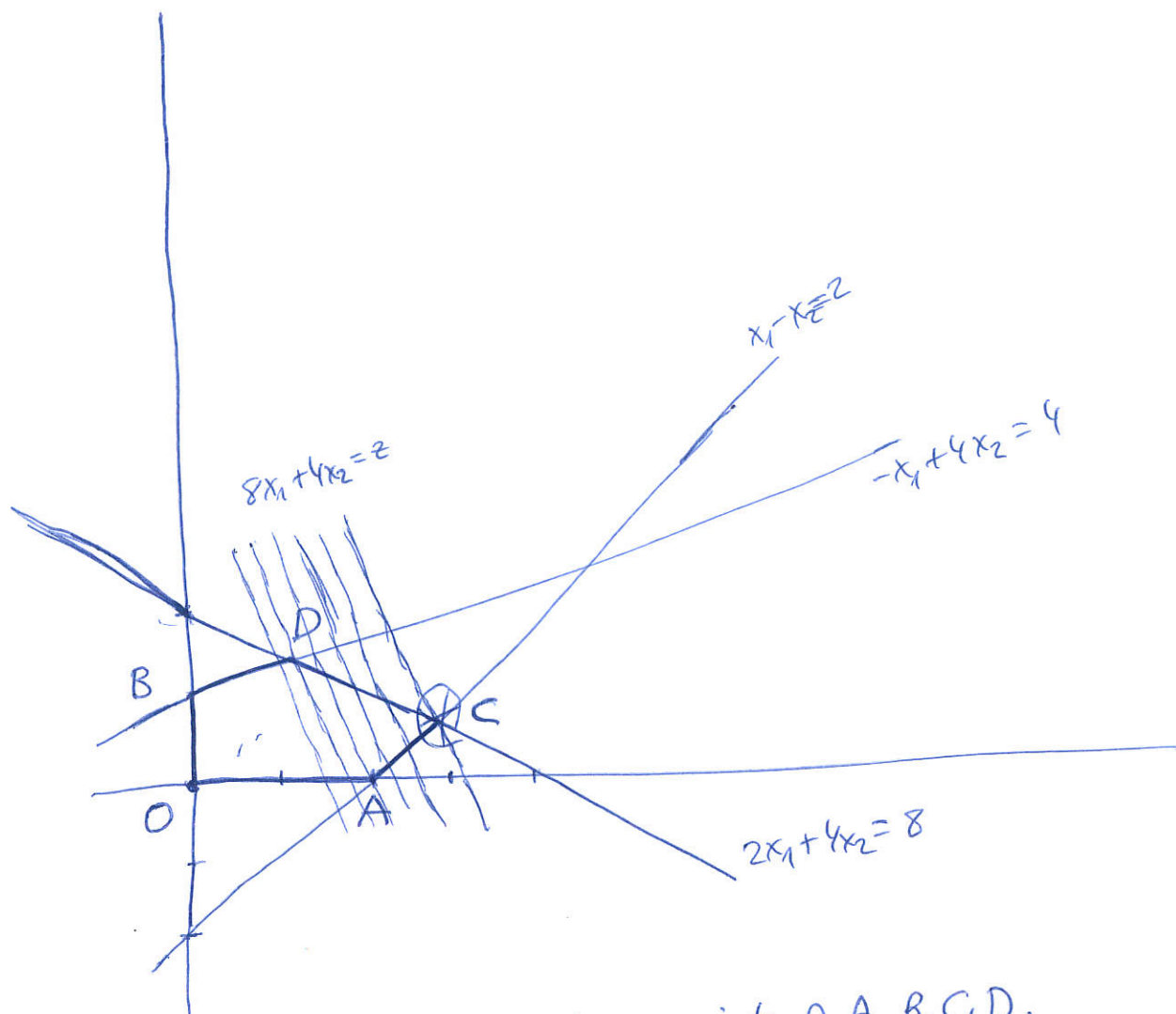
In our situation the Dual Problem is :

$$\max_{\lambda \in \mathbb{R}^5} - \begin{bmatrix} 8 \\ 4 \\ 2 \\ 0 \\ 0 \end{bmatrix}^T \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \\ \lambda_5 \end{bmatrix}$$

$$\text{subject to } \begin{cases} \begin{bmatrix} 2 & -1 & 1 & -1 & 0 \\ 4 & 4 & -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \\ \lambda_5 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix} \\ \lambda_i \geq 0 \quad (i=1,2,3,4,5) \end{cases}$$

(c) To solve the optimization problem using the graphical method, we need to make a graphic with all the lines involved in the constraints.

$$\begin{aligned} 2x_1 + 4x_2 &= 8 \rightarrow x_2 = -\frac{1}{2}x_1 + 2 \\ -x_1 + 4x_2 &= 4 \rightarrow x_2 = \frac{1}{4}x_1 + 1 \\ x_1 - x_2 &\leq 2 \rightarrow x_2 = x_1 - 2 \end{aligned}$$



The set S is the convex set with extreme points O, A, B, C, D .
 We find, equivalently, the point where $\max_{(x_1, x_2) \in S} 8x_1 + 4x_2$ is attained.
 Letting $8x_1 + 4x_2 = z$, we obtain parallel lines of slope -2 . The maximum value for z is obtained at point C , which is the intersection

$$\begin{cases} 2x_1 + 4x_2 = 8 \\ x_1 - x_2 = 2 \end{cases} \rightarrow (x_1, x_2) = \left(\frac{8}{3}, \frac{2}{3}\right). \text{ Hence } \boxed{C = \left(\frac{8}{3}, \frac{2}{3}\right)}$$
 is
 the minimum for our optimization problem, and the minimum
value of f on S is $-8 * \left(\frac{8}{3}\right) - 4 * \left(\frac{2}{3}\right) = -\frac{72}{3} = \boxed{-24}$.

Note that the dual problem can be stated, by solving $(*)$:

$$\begin{aligned} & \max -8\lambda_1 - 4\lambda_2 - 2\lambda_3 \\ \text{subject to } & \begin{cases} 2\lambda_1 - \lambda_2 + \lambda_3 \geq 8 \\ 4\lambda_1 + 4\lambda_2 - \lambda_3 \geq 4 \\ \lambda_1 \geq 0, \lambda_2 \geq 0, \lambda_3 \geq 0 \end{cases} \end{aligned}$$

The solution of this optimization problem is found in the intersection of the three planes

$$\begin{cases} 2\lambda_1 - \lambda_2 + \lambda_3 = 8 \\ 4\lambda_1 + 4\lambda_2 - \lambda_3 = 4 \\ \lambda_2 = 0 \end{cases}$$

which is the point $(\lambda_1, \lambda_2, \lambda_3) = (2, 0, 4)$.

Hence we see that the optimal value for the dual primal is

$$d^* = -8\lambda_1 - 4\lambda_2 - 2\lambda_3 = -8 \times 2 - 2 \times 4 = -24$$

Hence $d^* = -24 = p^*$, as predicted by the theory.

Short questions

(a)

$$\begin{aligned}\int_0^1 \int_0^x \int_0^{y-x^2} xyz \, dz dy dx &= \int_0^1 \int_0^x \left[xy \frac{z^2}{2} \right]_0^{y-x^2} dy dx \\&= \int_0^1 \int_0^x \frac{xy(y-x^2)^2}{2} dy dx = \frac{1}{2} \int_0^1 \int_0^x (xy^3 + x^5y - 2x^3y^2) dy dx \\&= \frac{1}{2} \int_0^1 \left[\frac{xy^4}{4} + \frac{x^5y^2}{2} - \frac{2x^3y^3}{3} \right]_0^x dx = \frac{1}{2} \int_0^1 \left[\frac{x^5}{4} + \frac{x^7}{2} - \frac{2x^6}{3} \right] dx \\&= \frac{1}{2} \left[\frac{x^6}{24} + \frac{x^8}{16} - \frac{2x^7}{21} \right]_0^1 = \frac{1}{2} \left[\frac{1}{24} + \frac{1}{16} - \frac{2}{21} \right] = \frac{1}{224}\end{aligned}$$

(b) $F(x,y) = (y, -x)$ acting on $\gamma(t) = (t^2, t^3)$, $0 \leq t \leq 1$.

$$\begin{aligned}W &= \int_0^1 F(\gamma(t)) \cdot \gamma'(t) dt = \\&= \int_0^1 (t^3, -t^2) \cdot (2t, 3t^2) dt = \\&= \int_0^1 [2t^4 - 3t^4] dt = \int_0^1 (-t^4) dt = \left[-\frac{t^5}{5} \right]_0^1 = -\frac{1}{5}.\end{aligned}$$